

BUCKLEY AFB, CO, USA

WMO: 724695

Lat: 39.717N

Lon: 104.750W

Elev: 5663

StdP: 11.93

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23036

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.5	8.1	-4.3	5.4	19.9	-0.5	6.6	24.4	26.7	36.6	24.0	39.6	6.6	170	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.0	93.4	58.5	90.9	58.6	88.4	58.5	64.3	78.1	63.1	77.9	62.0	77.1	8.6	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.1	99.6	66.3	59.2	92.9	66.2	57.4	86.8	66.0	32.4	77.3	31.4	77.4	30.5	77.3	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	20.7	18.1	DB	-5.3	98.7	4.8	2.2	-8.8	100.3	-11.5	101.5	-14.2	102.8	-17.7	104.3	
			WB	-5.6	67.6	4.7	1.5	-9.0	68.7	-11.8	69.5	-14.4	70.4	-17.9	71.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	33.3	34.6	42.0	47.6	56.6	67.4	74.1	71.9	64.4	51.4	40.1	33.6
	DBStd	17.27	10.81	11.48	10.63	9.49	8.62	7.02	5.00	4.83	8.58	10.03	11.30	10.40
	HDD50	2388	522	437	286	153	38	2	0	0	12	102	326	510
	HDD65	5668	983	851	714	521	279	53	4	8	109	426	748	972
	CDD50	2942	3	6	36	83	243	524	748	679	444	145	28	3
	CDD65	749	0	0	0	0	19	125	287	223	91	4	0	0
	CDH74	8843	0	1	9	34	384	1666	3170	2263	1171	143	2	0
	CDH80	3740	0	0	0	1	96	704	1543	962	418	16	0	0
Wind	WSAvg	8.9	8.7	9.2	10.0	10.4	9.3	8.8	8.4	8.2	8.2	8.4	8.5	8.7
Precipitation	PrecAvg	16.9	0.6	0.7	1.3	2.2	2.3	1.9	2.2	2.1	1.4	1.1	0.8	0.6
	PrecMax	24.1	1.3	2.2	4.3	5.5	5.9	4.4	4.4	4.5	6.9	2.8	2.7	2.8
	PrecMin	9.8	0.1	0.0	0.0	0.1	0.6	0.5	0.3	0.3	0.0	0.0	0.2	0.0
	PrecStd	3.4	0.3	0.5	1.0	1.2	1.3	1.1	1.2	1.1	1.3	0.7	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.5	68.0	75.0	77.5	88.3	94.9	97.4	95.1	92.5	82.0	72.9	64.2
		MCWB	43.4	43.8	47.2	49.4	56.3	57.6	59.0	58.6	57.8	51.9	46.1	42.2
	2%	DB	59.0	63.1	70.1	74.2	82.0	91.0	94.7	91.4	88.2	78.4	67.9	58.8
		MCWB	40.6	41.9	45.6	48.4	54.3	57.4	59.0	58.9	56.2	50.6	45.2	40.1
	5%	DB	54.2	57.8	65.5	71.0	78.6	88.1	91.5	89.1	85.3	73.9	63.3	54.1
		MCWB	38.4	40.0	43.8	47.3	53.3	57.4	59.2	59.1	55.8	49.3	43.7	38.1
	10%	DB	49.4	52.9	60.7	66.0	73.5	84.0	89.4	85.9	81.6	68.9	57.5	48.9
		MCWB	36.4	38.1	41.9	45.8	52.4	57.6	59.5	59.3	55.4	48.2	40.9	36.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	45.2	48.5	52.4	60.8	65.1	66.5	65.3	62.6	56.2	48.1	43.3
		MCDB	63.1	65.0	70.5	69.9	71.8	79.9	78.0	78.3	77.3	72.6	67.7	62.5
	2%	WB	41.2	42.5	46.4	50.2	57.8	62.4	64.7	63.7	60.1	53.1	45.9	40.7
		MCDB	56.6	60.7	67.2	68.9	73.4	76.6	77.6	79.3	76.4	68.4	65.8	57.4
	5%	WB	39.1	40.6	44.4	48.4	55.7	61.0	63.6	62.4	58.6	51.1	44.0	38.4
		MCDB	52.8	56.9	62.6	66.2	71.9	75.8	77.8	78.0	75.7	68.5	62.0	52.9
	10%	WB	36.7	38.5	42.5	46.8	54.0	59.5	62.4	61.4	57.2	49.4	41.7	36.2
		MCDB	48.3	52.0	58.7	63.4	70.0	75.0	77.4	76.8	74.4	66.6	57.0	48.5
Mean Daily Temperature Range	5% DB	MDBR	23.0	23.4	25.1	24.5	24.3	26.9	27.0	26.1	26.9	25.5	23.6	22.8
		MCDBR	28.3	30.0	31.5	31.6	30.7	32.0	30.8	30.5	31.1	31.8	30.6	28.2
		MCWBR	14.9	15.1	14.3	13.2	11.4	10.3	8.5	8.6	9.8	12.8	14.5	14.7
	5% WB	MCDBR	27.4	28.7	29.6	29.0	27.5	26.2	25.7	25.4	26.9	28.0	29.3	27.4
MCWBR		15.1	15.0	14.2	12.9	11.5	10.8	8.9	8.7	9.5	12.1	14.3	15.0	
Clear-Sky Solar Irradiance	taub	0.236	0.243	0.263	0.294	0.313	0.321	0.349	0.339	0.304	0.266	0.248	0.237	
	taud	2.631	2.625	2.571	2.492	2.501	2.490	2.452	2.467	2.546	2.642	2.613	2.619	
	Ebn at Noon	308	319	319	312	306	302	292	293	300	305	299	299	
	Edh at Noon	21	24	29	33	34	34	35	34	29	23	21	19	
All-Sky Solar Radiation	RadAvg	815	1125	1511	1801	2000	2279	2176	1947	1674	1231	884	714	
	RadStd	35	61	113	124	157	105	94	62	84	75	49	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	+0.54	+1.22	N/A	N/A	N/A	N/A
Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	N/A	N/A

Nomenclature: See separate page